

The Alien in the Desert

Concept Document

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Pitch Line

Among the dunes and fog, only the cleverest alien will escape! Navigate the treacherous desert, survive the sandstorms, and reach the UFO before it's too late.

Introduction

The Alien in the Desert is a tile-based game for 1 to 4 players. Each player controls an alien whose goal is to reach the center of the board, where a UFO awaits to rescue them from the desert.

The game features a fog-of-war mechanic, which limits players' vision to only the next two tiles ahead. Consequently, when a player rolls the die, they can choose to move either the number of tiles they can see or the full number rolled.

Some tiles contain a tornado. If an alien lands on one, they lose 1 life point. Each alien starts with 2 life points; if they lose both, they are eliminated from the game.

Demographic Breakdown

Due to its simple controls and mechanics, this game is recommended for ages 8 and up and is only available on PC.

Feature List

- Tile-based board game for 1–4 players.
- Each player controls an alien trying to reach the spaceship at the center of the board.
- "Fog of War" system limits visibility to two tiles ahead.
- Players roll a die to determine movement distance each turn.
- Choice mechanic: move only within visible range or risk full die result.
- Randomly placed tiles create a unique desert map each game.
- Tornado tiles cause 1 point of damage when landed on.
- Each alien starts with 2 life points; reaching 0 eliminates the player.

- The first alien to reach the spaceship wins the game.
- Desert survival theme with strategic movement and risk–reward gameplay.

Feature List Breakdown

- **Tile-Based Exploration Gameplay:** *The Alien in the Desert* is a tile-based adventure game focused on exploration, risk management, and survival. Players must navigate a shifting desert map to reach the center and escape.
- **Control an Alien Explorer:** Each player controls a unique alien character stranded in a vast desert. The goal is to reach the spaceship at the center of the board before anyone else — or before losing all life points.
- **1–4 Player Modes:** The game supports 1 to 4 players, offering both solo and local multiplayer experiences. In solo mode, players can test their strategy and adaptability against the unpredictable desert. In multiplayer, players compete to be the first to reach the spaceship.
- **Fog of War Visibility System:** The “fog of war” mechanic limits vision to only two tiles ahead. This creates tension and uncertainty, forcing players to make decisions with limited information about what lies ahead.
- **Dice-Based Movement with Risk–Reward Choice:** Each turn, players roll a die to determine movement distance. After rolling, they must choose between moving only within their visible range (safe) or using the full die result (risky), potentially encountering hidden hazards.
- **Randomized Tile Layout:** At the start of each game, tiles are shuffled and placed randomly, ensuring that no two matches are ever the same. This randomness keeps the gameplay fresh and unpredictable.
- **Tornado Hazard Tiles:** Some tiles contain tornadoes that damage any alien landing on them. Each tornado hit costs the player 1 life point and forces them to reassess their route through the desert.
- **Health and Elimination System:** Every alien starts with 2 life points. If an alien’s health drops to zero, it is eliminated from the game. Players must balance bold exploration with self-preservation.
- **Win Condition:** The first alien to reach the spaceship in the center of the board wins. If all players are eliminated before anyone arrives, the desert claims victory — and no one escapes.
- **Desert Survival Theme and Style:** The game’s atmosphere combines mystery, isolation, and danger, set in a vast, shifting alien desert. The visual and thematic design emphasizes survival, strategy, and the thrill of the unknown.

Formal Description

Game State:

The game state contains all relevant information at a specific moment in time:

- `player_positions`: current board positions of each player (`Array[int]`);
- `player_lives`: remaining life points for each player (`Array[float]`);
- `current_turn`: index of the player currently taking their turn (`int`);
- `current_roll_value`: result of the latest dice roll (`int`);
- `FINAL_SLOT`: final slot on the board (`int`, constant = 37);
- `hurricanes_positions`: map positions occupied by hurricanes (`Array[int]` per player);
- `active_players`: number of players currently active (`Global.num_players`) (`int`);
- `game_over`: indicates whether the game has ended (`boolean`).

`S={player_positions,player_lives,current_turn,current_roll_value,FINAL_SLOT,hurricanes_positions,active_players,game_over}`

Game View:

What each player can see on the screen:

- Their own current position on the map (`player_positions[i]`);
- Their life bar (`player_lives[i]`);
- The rolled dice face (`current_roll_value`);
- The active player indicator (`current_turn`);
- Available action buttons ("Roll Dice", "Advance 2", "Advance X");
- Visual effects such as hurricanes and fog of war.

Players take turns; only one player acts at a time while others observe.

Game Setup:

Initial setup:

- `player_positions` = `[0, 0, 0, 0]`
- `player_lives` = `[2.0, 2.0, 2.0, 2.0]`
- `current_turn` = 0

- Hurricanes are randomly placed for each player (HURRICANES_PER_PLAYER = 7).
- All UI buttons and sprites are hidden until the first turn begins.

S0={player_positions_i=0,player_lives_i=2.0,current_turn=0}

Player Actions:

Actions available to the active player during their turn:

- **Roll Dice:** generate a random value between 1 and 6 (`_on_button_pressed()`), stored in `current_roll_value`;
- **Advance 2 Spaces:** move the player 2 slots forward (`_on_advance2_pressed()`);
- **Advance X Spaces:** move the player forward by the rolled dice value (`_on_advancex_pressed()`);
- **Take Damage:** lose 1 life point when colliding with a hurricane (`check_for_hurricane_hit()`).

A={roll_dice(),advance(steps),take_damage(),skip_turn()}

Victory Conditions:

- Victory: A player wins upon reaching or surpassing FINAL_SLOT (the end of the board).
- Defeat: The game ends if all players have `player_lives <= 0`, resulting in no winner.
- Transition: When the game concludes, the scene switches to `Victory.tscn`, displaying the winner or a tie outcome.

$V = (\text{player_positions}[i] \geq \text{FINAL_SLOT}) \Rightarrow \text{Victory}(i)$

Progression of Play:

Each turn proceeds as follows:

1. The active player rolls the dice.
2. The player chooses to advance **2 spaces** or **X spaces**, based on the dice result.
3. The character moves step-by-step along the path (`move_player()`), updating their position.
4. If a hurricane is encountered, the player loses 1 life point.

5. If a player's life reaches 0, they are eliminated.
6. If a player reaches or surpasses the final slot (FINAL_SLOT), they win.
7. Otherwise, the game advances to the next living player, and the cycle repeats.